

$$\begin{aligned}
& \tilde{\mu} \, \tilde{g}_1^4 \, y_d^{i1i2} \, \text{LF}_{2,2,0}[\mathbf{m}_1, \tilde{\mu}] - \frac{1}{12} \, m_2 \, \tilde{\mu} \, \tilde{g}_2^4 \, (9 \, y_d^{i1i2} + 4 \, y_d^{p12} \, \delta_{p11}) \, \text{LF}_{2,2,0}[\mathbf{m}_2, \tilde{\mu}] + \\
& \tilde{g}_2^4 \, y_d^{p12} \, \text{LF}_{3,2,-1}[\mathbf{m}_2, \tilde{\mu}] \, \delta_{p11} + \frac{1}{18} \, \tilde{\mu} \, \tilde{g}_1^2 \, y_d^{pR} \, \overline{\mathbf{a}}_d^{pR} \, (-3 \, y_d^{i1i2} + y_d^{s12} \, \delta_{s11} + 2 \, y_d^{i1s} \, \delta_{s12}) \\
& \times [\mathbf{m}_d^r, \mathbf{m}_q^p] + \frac{1}{36} \, \tilde{\mu} \, \tilde{g}_1^2 \, y_d^{pR} \, \overline{\mathbf{a}}_d^{pR} \, (3 \, y_d^{i1i2} - y_d^{s12} \, \delta_{s11} - 2 \, y_d^{i1s} \, \delta_{s12}) \, \text{LF}_{3,2,-1}[\mathbf{m}_d^r, \mathbf{m}_q^p] + \\
& \tilde{g}_2^2 \, y_d^{pR} \, \overline{\mathbf{a}}_e^{pR} \, (-3 \, y_d^{i1i2} + y_d^{s12} \, \delta_{s11} + 2 \, y_d^{i1s} \, \delta_{s12}) \, \text{LF}_{2,2,0}[\mathbf{m}_e^r, \mathbf{m}_l^p] + \\
& \tilde{g}_2^2 \, y_d^{pR} \, \overline{\mathbf{a}}_e^{pR} \, (3 \, y_d^{i1i2} - y_d^{s12} \, \delta_{s11} - 2 \, y_d^{i1s} \, \delta_{s12}) \, \text{LF}_{3,2,-1}[\mathbf{m}_e^r, \mathbf{m}_l^p] + \\
& \tilde{g}_2^2 \, y_d^{pR} \, \overline{\mathbf{a}}_e^{pR} \, (3 \, y_d^{i1i2} - y_d^{s12} \, \delta_{s11} - 2 \, y_d^{i1s} \, \delta_{s12}) \, \text{LF}_{3,1,0}[\mathbf{m}_l^p, \mathbf{m}_e^r] + \\
& \tilde{g}_2^2 \, y_d^{pR} \, \overline{\mathbf{a}}_e^{pR} \, (3 \, y_d^{i1i2} - y_d^{s12} \, \delta_{s11} - 2 \, y_d^{i1s} \, \delta_{s12}) \, \text{LF}_{3,2,-1}[\mathbf{m}_l^p, \mathbf{m}_e^r] + \frac{1}{8} \, \tilde{\mu} \, y_e^{pR} \, \overline{\mathbf{a}}_e^{pR} \\
& (7 \, y_d^{i1i2} + y_d^{s12} \, \delta_{s11}) + \mathbf{g}_1^2 \, (-3 \, y_d^{i1i2} + y_d^{s12} \, \delta_{s11} + 2 \, y_d^{i1s} \, \delta_{s12}) \, \text{LF}_{4,1,-1}[\mathbf{m}_l^p, \mathbf{m}_e^r] + \\
& \tilde{g}_1^2 \, y_d^{pR} \, \overline{\mathbf{a}}_e^{pR} \, (3 \, \mathbf{g}_2^2 \, (6 \, y_d^{i1i2} + y_d^{s12} \, \delta_{s11}) + \mathbf{g}_1^2 \, (3 \, y_d^{i1i2} - y_d^{s12} \, \delta_{s11} - 2 \, y_d^{i1s} \, \delta_{s12})) \\
& \times [\mathbf{m}_l^p, \mathbf{m}_e^r] + \frac{1}{18} \, \tilde{\mu} \, \tilde{g}_1^2 \, y_d^{pR} \, \overline{\mathbf{a}}_d^{pR} \, (3 \, y_d^{i1i2} - y_d^{s12} \, \delta_{s11} - 2 \, y_d^{i1s} \, \delta_{s12}) \, \text{LF}_{3,1,0}[\mathbf{m}_q^p, \mathbf{m}_d^r] + \\
& \tilde{g}_1^2 \, y_d^{pR} \, \overline{\mathbf{a}}_d^{pR} \, (3 \, y_d^{i1i2} - y_d^{s12} \, \delta_{s11} - 2 \, y_d^{i1s} \, \delta_{s12}) \, \text{LF}_{3,2,-1}[\mathbf{m}_q^p, \mathbf{m}_d^r] + \\
& \tilde{g}_1^2 \, y_d^{pR} \, \overline{\mathbf{a}}_d^{pR} \, (-9 \, \mathbf{g}_2^2 \, (7 \, y_d^{i1i2} + y_d^{s12} \, \delta_{s11}) + 5 \, \mathbf{g}_1^2 \, (-3 \, y_d^{i1i2} + y_d^{s12} \, \delta_{s11} + 2 \, y_d^{i1s} \, \delta_{s12})) \\
& \times [\mathbf{m}_q^p, \mathbf{m}_d^r] + \frac{1}{6} \, \tilde{\mu} \, y_d^{pR} \, \overline{\mathbf{a}}_d^{pR} \\
& (6 \, y_d^{i1i2} + y_d^{s12} \, \delta_{s11}) + \mathbf{g}_1^2 \, (3 \, y_d^{i1i2} - y_d^{s12} \, \delta_{s11} - 2 \, y_d^{i1s} \, \delta_{s12}) \, \text{LF}_{5,1,-2}[\mathbf{m}_q^p, \mathbf{m}_d^r] + \\
& \tilde{g}_1^2 \, y_d^{pR} \, \overline{\mathbf{a}}_d^{pR} \, ((\mathbf{g}_1^2 + 9 \, \mathbf{g}_2^2) \, (3 \, y_d^{i1i2} - y_d^{s12} \, \delta_{s11}) - 2 \, \mathbf{g}_1^2 \, y_d^{i1s} \, \delta_{s12}) \, \text{LF}_{2,2,0}[\mathbf{m}_q^p, \mathbf{m}_u^r] + \\
& \tilde{g}_2^2 \, y_d^{i1i2} \, y_u^{pR} \, \overline{\mathbf{a}}_d^{pR} \, \text{LF}_{3,1,0}[\mathbf{m}_l^p, \mathbf{m}_u^r] + \\
& \tilde{g}_1^2 \, y_d^{pR} \, \overline{\mathbf{a}}_d^{pR} \, (-\mathbf{g}_1^2 + 9 \, \mathbf{g}_2^2) \, (3 \, y_d^{i1i2} - y_d^{s12} \, \delta_{s11}) + 2 \, \mathbf{g}_1^2 \, y_d^{i1s} \, \delta_{s12} \, \text{LF}_{3,2,-1}[\mathbf{m}_l^p, \mathbf{m}_u^r] - \\
& \tilde{g}_2^2 \, y_d^{i1i2} \, y_u^{pR} \, \overline{\mathbf{a}}_d^{pR} \, \text{LF}_{4,1,-1}[\mathbf{m}_q^p, \mathbf{m}_u^r] + \frac{1}{36} \, \tilde{\mu} \, y_u^{pR} \, \overline{\mathbf{a}}_d^{pR} \\
& (6 \, y_d^{i1i2} + y_d^{s12} \, \delta_{s11}) + \mathbf{g}_1^2 \, (-3 \, y_d^{i1i2} + y_d^{s12} \, \delta_{s11} + 2 \, y_d^{i1s} \, \delta_{s12}) \, \text{LF}_{3,1,0}[\mathbf{m}_l^r, \mathbf{m}_q^p] + \\
& \tilde{g}_1^2 \, y_d^{pR} \, \overline{\mathbf{a}}_d^{pR} \, (-\mathbf{g}_1^2 + 9 \, \mathbf{g}_2^2) \, (3 \, y_d^{i1i2} - y_d^{s12} \, \delta_{s11}) + 2 \, \mathbf{g}_1^2 \, y_d^{i1s} \, \delta_{s12} \, \text{LF}_{3,2,-1}[\mathbf{m}_l^r, \mathbf{m}_q^p] + \\
& \tilde{g}_1^2 \, y_d^{pR} \, \overline{\mathbf{a}}_d^{pR} \, (-9 \, \mathbf{g}_2^2 \, (6 \, y_d^{i1i2} + y_d^{s12} \, \delta_{s11}) + \mathbf{g}_1^2 \, (-3 \, y_d^{i1i2} + y_d^{s12} \, \delta_{s11} + 2 \, y_d^{i1s} \, \delta_{s12})) \\
& \times [\mathbf{m}_l^r, \mathbf{m}_q^p] + \frac{1}{6} \, \tilde{\mu} \, y_u^{pR} \, \overline{\mathbf{a}}_d^{pR} \\
& (6 \, y_d^{i1i2} + y_d^{s12} \, \delta_{s11}) + \mathbf{g}_1^2 \, (3 \, y_d^{i1i2} - y_d^{s12} \, \delta_{s11} - 2 \, y_d^{i1s} \, \delta_{s12}) \, \text{LF}_{5,1,-2}[\mathbf{m}_l^r, \mathbf{m}_q^p] + \\
& \mathbf{g}_1^2 \, \mathbf{g}_2^2 \, y_d^{i1i2} \, \text{LF}_{3,1,0}[\tilde{\mu}, \mathbf{m}_1] + \frac{7}{8} \, m_1 \, \tilde{\mu} \, \tilde{g}_1^4 \, y_d^{i1i2} \, \text{LF}_{3,2,-1}[\tilde{\mu}, \mathbf{m}_1] + \\
& \tilde{\mu} \, \mathbf{g}_1^2 \, (-3 \, \mathbf{g}_2^2 \, (12 \, y_d^{i1i2} + y_d^{p12} \, \delta_{p11}) + \mathbf{g}_1^2 \, (-3 \, y_d^{i1i2} + y_d^{p12} \, \delta_{p11} + 2 \, y_d^{i1p} \, \delta_{p12})) \\
& \times [\tilde{\mu}, \mathbf{m}_1] - \frac{1}{2} \, m_1 \, \tilde{\mu} \, \tilde{g}_1^4 \, y_d^{i1i2} \, \text{LF}_{4,2,-2}[\tilde{\mu}, \mathbf{m}_1] + \\
& \tilde{\mu} \, \mathbf{g}_1^2 \, (3 \, \mathbf{g}_2^2 \, (6 \, y_d^{i1i2} + y_d^{p12} \, \delta_{p11}) + \mathbf{g}_1^2 \, (3 \, y_d^{i1i2} - y_d^{p12} \, \delta_{p11} - 2 \, y_d^{i1p} \, \delta_{p12})) \\
& \times [\tilde{\mu}, \mathbf{m}_1] + \frac{1}{6} \, m_2 \, \tilde{\mu} \, \tilde{g}_2^4 \, (9 \, y_d^{i1i2} + 2 \, y_d^{p12} \, \delta_{p11}) \, \text{LF}_{3,1,0}[\tilde{\mu}, \mathbf{m}_2] + \\
& \tilde{\mu} \, \mathbf{g}_2^4 \, (57 \, y_d^{i1i2} + 8 \, y_d^{p12} \, \delta_{p11}) \, \text{LF}_{3,2,-1}[\tilde{\mu}, \mathbf{m}_2] + \\
& \mathbf{g}_2^2 \, (-\mathbf{g}_2^2 \, (36 \, y_d^{i1i2} + 7 \, y_d^{p12} \, \delta_{p11}) + \mathbf{g}_1^2 \, (-3 \, y_d^{i1i2} + y_d^{p12} \, \delta_{p11} + 2 \, y_d^{i1p} \, \delta_{p12})) \\
& \times [\tilde{\mu}, \mathbf{m}_2] - 2 \, m_2 \, \tilde{\mu} \, \tilde{g}_2^4 \, y_d^{i1i2} \, \text{LF}_{4,2,-2}[\tilde{\mu}, \mathbf{m}_2] + \\
& \mathbf{g}_2^2 \, (3 \, \mathbf{g}_2^2 \, (6 \, y_d^{i1i2} + y_d^{p12} \, \delta_{p11}) + \mathbf{g}_1^2 \, (3 \, y_d^{i1i2} - y_d^{p12} \, \delta_{p11} - 2 \$$